DV2626/DV2599 Machine Learning

Lecture 9: Unsupervised Learning and Clustering

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Outline

- Supervised vs unsupervised learning
- Clustering overview and examples
  - Clustering goal and representation
  - Predictive vs descriptive clustering
  - Hard vs soft clustering
  - Flat vs hierarchical clustering
  - Examples
  - …
- Evaluation of clustering solutions
- References

Supervised vs unsupervised learning

- In supervised learning scenarios we have a labeled training set
  - learning conditional distribution $P(Y|X)$, X: features, Y: classes

- In unsupervised learning we do not have access to class-labels
  - learning distribution $P(X)$, X: features

- Unsupervised learning settings
  - Predictive clustering - trying to predict the cluster an instance belongs to
  - Descriptive clustering - clustering data to discover how instances are connected

Predictive vs descriptive clustering

- Predictive clustering
  - similar to classification
  - learn a clustering model from training data that can subsequently be used to assign new data to clusters
  - distinction between the task (clustering arbitrary data) and the learning problem (learning a clustering model from training data)

- Descriptive clustering
  - explorative clustering
  - extract knowledge about the data
  - can not be used to predict new instance
  - can be used for analysis, organization (finding a structure) and understanding of data

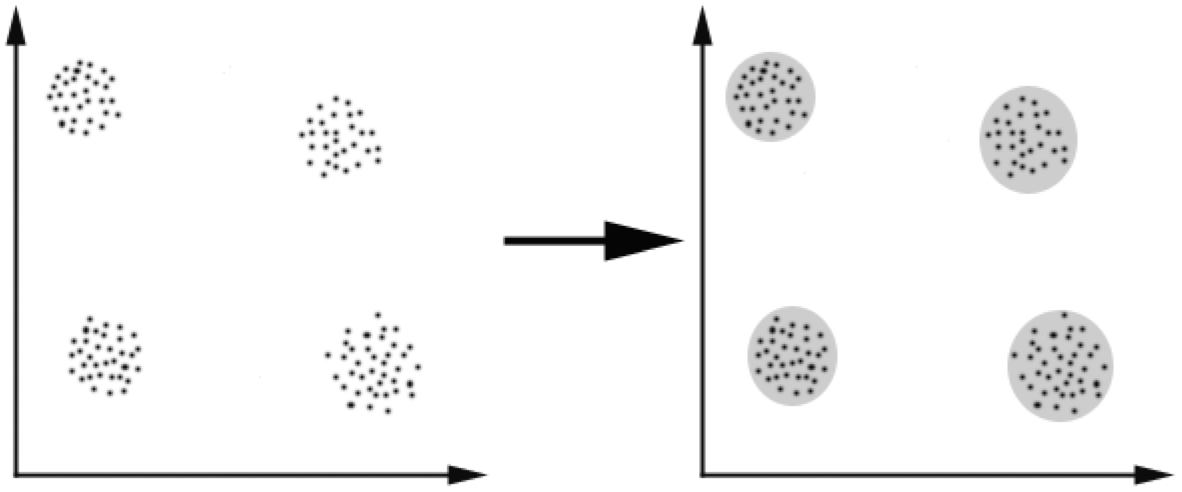
Hard vs soft clustering

- Hard clustering
  - same instance can only belong to single cluster
  - the generated clusters are disjoint, i.e. they have an empty intersection

- Soft clustering
  - same instance can belong to different clusters
  - each instance can be associated with a degree to which it belongs to a cluster
  - the clusters are not disjoint

What is clustering?

Find *k* clusters (or a classification that consists of *k* clusters) so that the objects of one cluster are similar to each other whereas objects of different clusters are dissimilar - Bacher (1996)



Settings ⇨ Objective function ⇨ Algorithm

Clustering goal

- Determine the intrinsic grouping in a set of unlabeled data.

- What constitutes a good clustering?

- All clustering algorithms will produce clusters, regardless of whether the data contains them

- There is no golden standard, depends on goal:
  - data reduction
  - “natural clusters”
  - “useful” clusters
  - outlier detection

How to measure similarity?

- Similarity is a fundamental concept in theories of knowledge and behavior
- Psychological experiments have shown that similarity acts as an organizing principle by which individuals classify objects, and make generalisations – Goldstone (1994)
- We usually use distance (dissimilarity) to compare two instances
- The suitable distance function is dependent on data and problem under consideration
  - Euclidian distance, Jaccard distance, Manhattan distance, Dynamic Time Warping distance, …

Representation of clustering models

- The representation of clustering models depends on whether they are predictive, descriptive or soft.

- A descriptive clustering of n data points into c clusters could be represented by a partition matrix: an n-by-c binary matrix with exactly one 1 in each row.

- A soft clustering corresponds to a row-normalised n-by-c matrix.

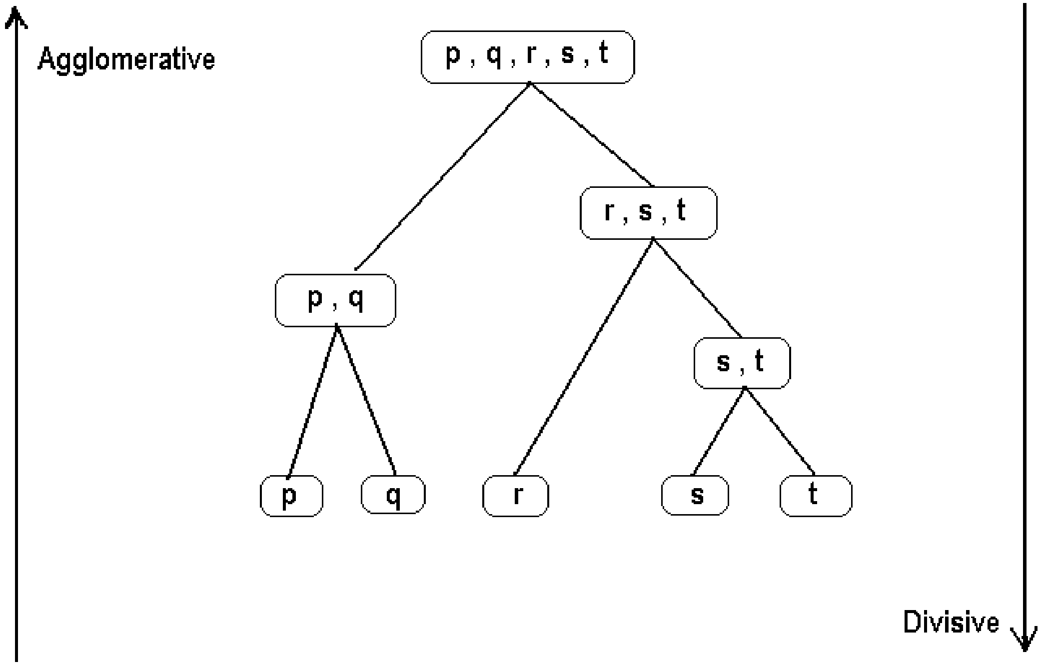
- Predictive clustering methods represent a cluster by its centroid or exemplar.

Flat vs hierarchical clustering

- Flat clustering (partitional)
  - clusters are flat
- Hierarchical clustering
  - clusters form a tree
  - agglomerative (bottom-up) and divisive (top-down)

Hierarchical clustering

- Agglomerative clustering treats each data point as a singleton cluster, and then successively merges clusters until all points have been merged into a single remaining cluster.
- Divisive clustering works the other way around.

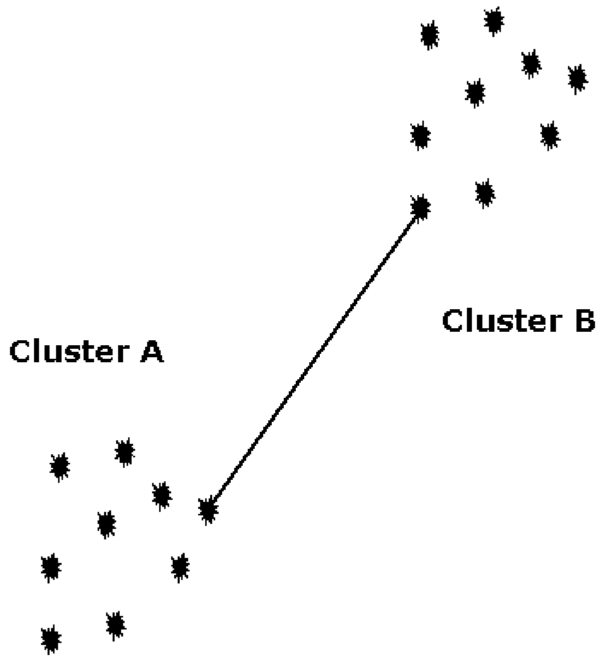


Agglomerative clustering (1)

- Place each of *n* instances into a class of its own
- Compute all pair-wise instance-instance similarity scores
- Merge the two most similar clusters into one
  - Replace the two clusters into the new cluster
  - Re-compute inter-cluster similarity scores w.r.t. the new clusters
- Repeat the above step until there are *k* clusters left (*k* can be 1)

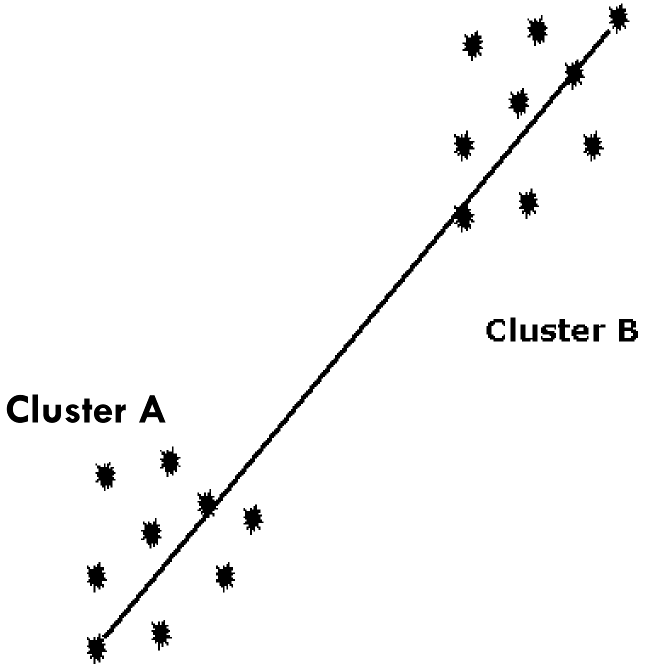
Agglomerative clustering (2)

In single-link hierarchical clustering, we merge in each step the two clusters whose two closest members have the smallest distance.



Agglomerative clustering (3)

In complete-link hierarchical clustering, we merge in each step the two clusters whose merger has the smallest diameter.



Average linkage is a compromise between the sensitivity of complete-link clustering to outliers and the tendency of single-link clustering to form long chains that do not correspond to the intuitive notion of clusters as compact, spherical objects.

Divisive clustering

- Start at the top with all instances in one cluster
- The cluster is split using a partitional clustering algorithm
- This procedure is applied recursively until each instance is in its own singleton cluster

Agglomerative vs divisive clustering

- Divisive is more complex, because a flat clustering is needed as a “subroutine”
- For a fixed number of top levels, using an efficient flat algorithm like *K*-means, divisive algorithms are linear in the number of instances and clusters
- Agglomerative algorithms are least quadratic
- Agglomerative methods make clustering decisions based on local patterns without initially taking into account the global distribution. These early decisions cannot be undone
- Divisive clustering benefits from complete information about the global distribution when making top-level partitioning decisions

Hybrid hierarchical clustering

- Bottom-up clustering is good at identifying small clusters, but can provide sub-optimal performance for identifying a few large clusters.

- Top-down methods are good at identifying a few large clusters, but weaker at many small clusters

- Hybrid clustering methods can combine the strengths of bottom-up hierarchical clustering with that of top-down clustering

K-means clustering (1)

- *K*-means clustering is a partitioning algorithm
  - Decompose the data set into ***k*** **disjoint clusters minimizing** the within-cluster sum of distances
  - The cluster center is the mean data vector averaged over all objects in the cluster

K-means clustering (2)

1. Start with a random initialization of *K* cluster centers

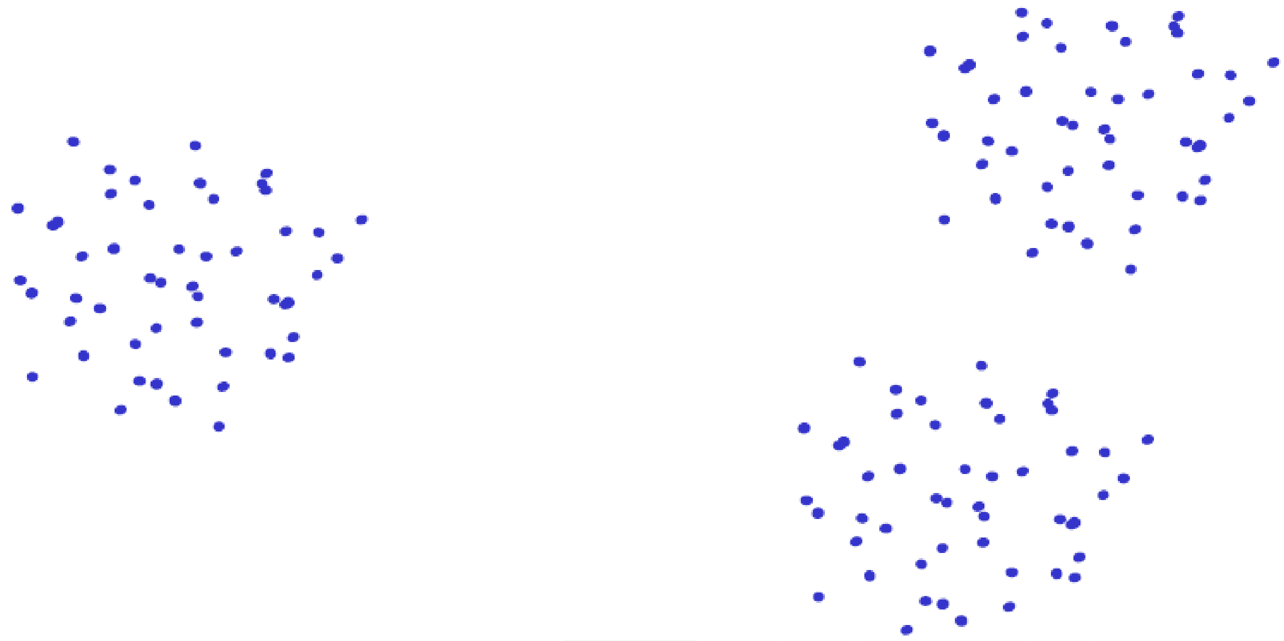
2. Generate a partition by assigning each pattern (data point) to its closest cluster center

3. Compute new cluster centers as the centroids of the clusters.

4. Steps 2 and 3 are repeated until there is no change in the membership (also cluster centers remain the same)

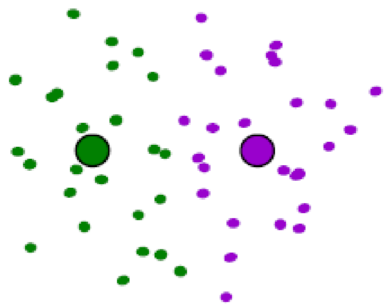
K-means clustering (3)

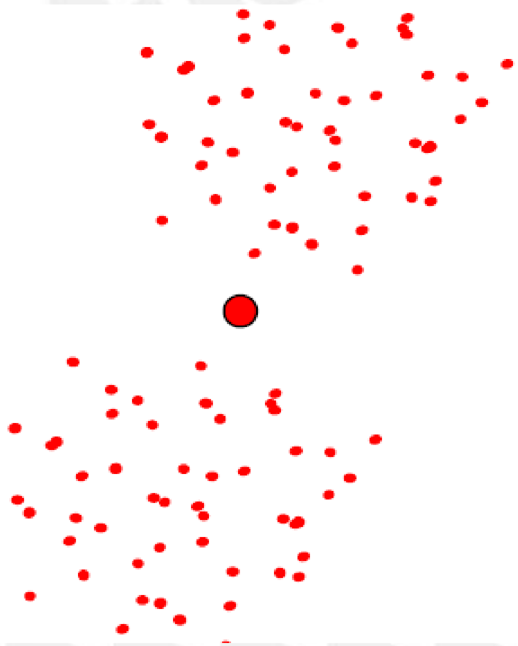
- Distavantages
  - Dependent on initialization



K-means clustering (4)

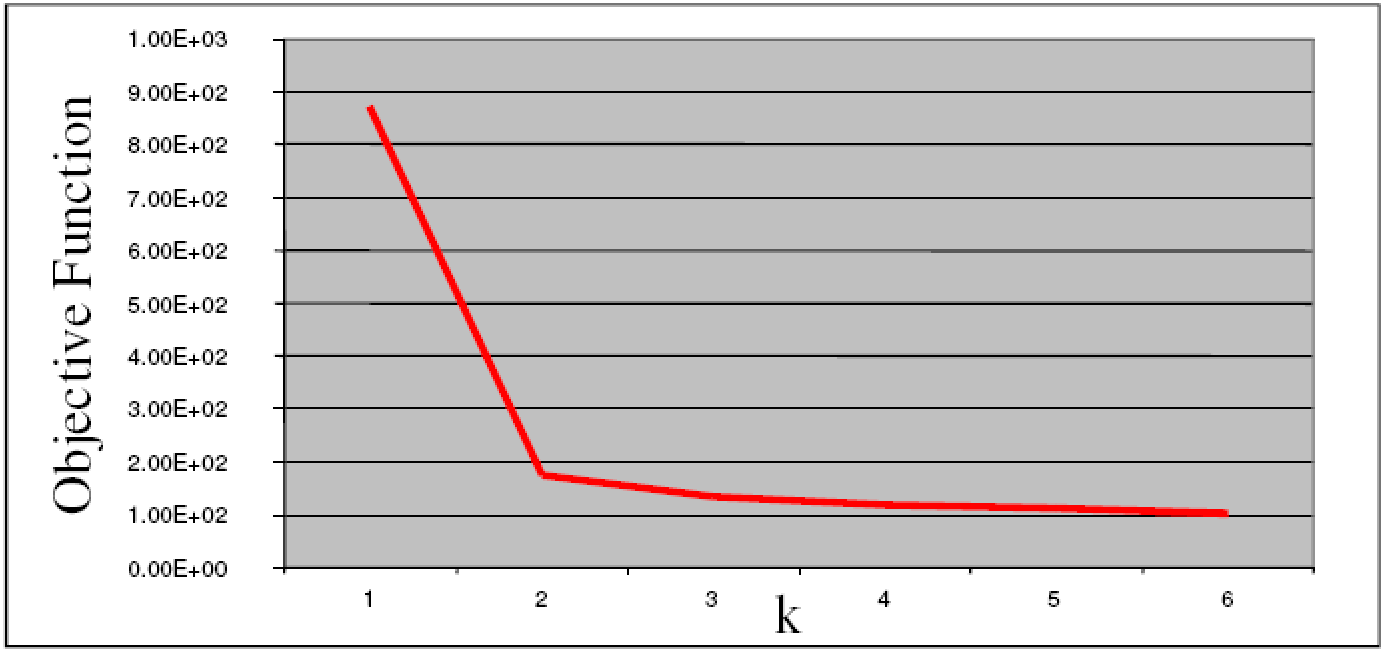
- Distavantages
  - Dependent on initialization
  - Sensitive to outliers (*K*-medoids)
  - Can deal only with clusters with spherical symmetrical point distribution





K-means clustering (5)

- Distavantages
  - Dependent on initialization
  - Sensitive to outliers (K-medoids)
  - Can deal only with clusters with spherical symmetrical point distribution
  - Determining *K* in advance



K-modes clustering

- *K-modes* is an extension of *K-means* for categorical data
  - Instead of distances it uses *dissimilarities, i.e.* quantification of the total mismatches between two objects: the smaller this number, the more similar the two objects
  - Instead of means, it uses *modes*
  - A mode is a vector of elements that minimizes the dissimilarities between the vector itself and each object of the cluster.

- *K-prototypes is another extension combing K-means* and *K-modes for* numerical and categorical data

Graph-based clustering: Minimum Cut Tree (1)

- The cut-clustering algorithm is a graph-based clustering algorithm
- The input data is represented by a similarity (an undirected) graph
  - each node is a data point and two nodes are connected if the similarity between the corresponding data points is positive, and the edge is weighted by the corresponding similarity score.
- The algorithm works by adding an artificial node to the existing graph and connected to all nodes in the graph with a value α.
- A minimum cut tree is computed and the artificial node is removed.
- The clusters consist of the nodes connected after the artificial node has been removed.

Graph-based clustering: Minimum Cut Tree (2)

Algorithm. Cut clustering algorithm[1]. Input is a connected weighted graph (G) with nodes (V), edges (E) and edge weight function w.

1: **function** Cut Clustering $G(V, E, w)$, α
2: $Vt = V \cup \{t\}$
3: **for** all nodes $v \in V$ **do**
4: Connect t to v with edge weight α
5: **end for**
6: $Gt = (Vt, Et, wt)$ is the expanded graph after connecting t to V
7: Calculate the Min-cut Tree T of Gt
8: Remove t from T
9: **return** all connected components as clusters of G
10: **end function**

[1]Flake G.W., R. E. Tarjan, and K. Tsioutsiouliklis, Graph Clustering and Minimum Cut Trees, Internet Mathematics, 1(4), 2004, p. 385-408.

Other clustering algorithms

- Spectral clustering
  - a graph-based clustering algorithm that uses similarity matrixes and eigenvalues to reduce features before clustering
- Expectation-maximisation clustering
  - a soft clustering algorithm that is based on the probability (expectation) of a data point belonging to a particular cluster
- Density based clustering algorithms (DBSCAN)
  - clusters are assumed as regions in the data space in which the objects are dense and the clusters are separated by regions of low object density
  - these regions have an arbitrary shape and the data points inside a cluster may be arbitrarily distributed
- Bipartite correlation clustering
  - a bipartite graph is given as input, and a set of disjoint clusters covering the graph nodes is output
- …

Cluster Validity

- For cluster analysis, the question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
- Why do we want to evaluate them?
  - To avoid finding patterns in noise
  - To compare clustering algorithms
  - To compare two sets of clusters (clustering solutions)
  - To compare two clusters
- The procedure of evaluating the results of a clustering algorithm is known under the term cluster validity

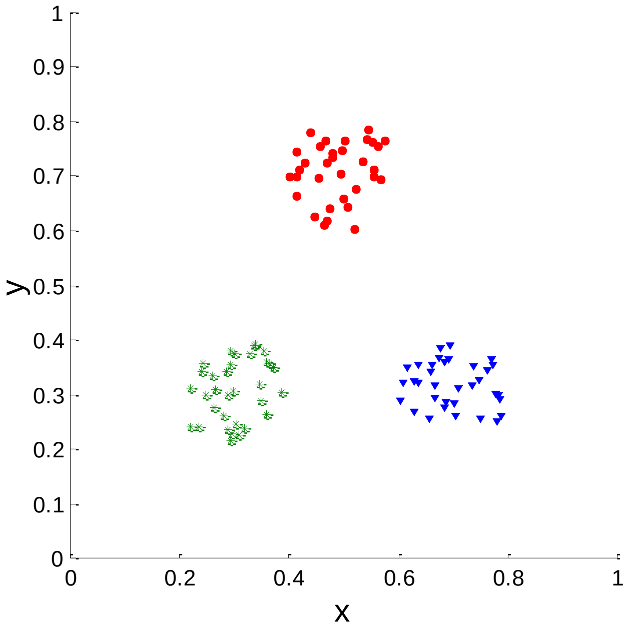
Different Aspects of Cluster Validation

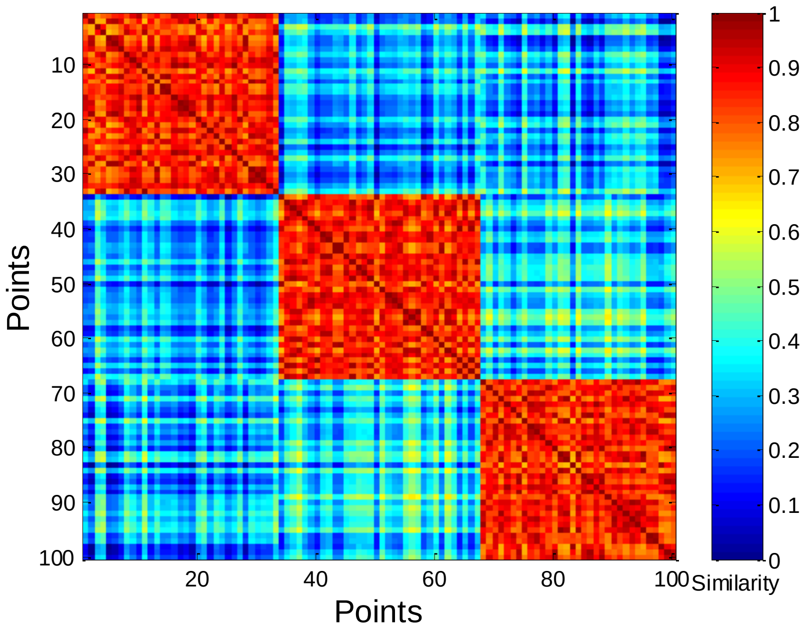
1. Determining the clustering tendency of a set of data, i.e., distinguishing whether non-random structure actually exists in the data.
2. Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.
3. Evaluating how well the results of a cluster analysis fit the data without reference to external information.
   - Use only the data
4. Comparing the results of two different sets of cluster analyses to determine which is better.
5. Determining the ‘correct’ number of clusters.

For 2, 3, and 4, we can further distinguish whether we want to evaluate the entire clustering or just individual clusters.

Cluster validation using similarity matrix

- Order the similarity matrix with respect to cluster labels and inspect visually.





Cluster Validation Measures

- External Measures
  - an independent assessment of clustering quality with respect to a pre-specified structure
- Internal Measures
  - base their validation on the same information used to derive the clusters themselves
- Relative Measures
  - evaluate the clustering result by comparing clustering solutions generated by the same algorithm but with different input parameter values

External Cluster Validation Indices

- *Unary external evaluation measures* take a single clustering result as the input, and compare it with a known set of class labels to assess the degree of consensus between the two.
  - An example of unary measure is the *F*-measure.

- *Symmetric measures* assess the consensus between a produced partitioning and the existing one based on the contingency table of the pairwise assignment of data items.

- Symmetric measures can be used as *binary* measures, *i.e.*, for assessing the similarity of two different clustering results.
  - the best known such index is the *Rand Index* (same as accuracy)

Internal Cluster Validation Indices

- Internal Indices can reflect different clustering properties:
  - *Compactness* evaluates the cluster homogeneity that is related to the closeness within a given cluster.
  - *Separation* demonstrates the opposite trend by assessing the degree of separation between individual groups.
  - *Connectedness* quantifies to what extent the nearest neighboring data items are placed into the same cluster.
  - The *stability* measures evaluate the consistency of a given clustering partition by clustering from all but one attribute (feature). The remaining attribute is subsequently used to assess the predictive power of the resulting clusters by measuring the within-cluster similarity in removed attribute.

Internal Measures: *Silhouette Index*

- Silhouette Index combines ideas of both compactness and separation for individual clusters, as well as clustering solutions

- For an individual point, *i*
  - Calculate a_i = average distance of *i* to the points in its cluster
  - Calculate b_i = min (average distances of *i* to points in another clusters)
  - The silhouette index for point *i* is then given by

    $s_i = (b_i - a_i)/\max\{a_i, b_i\}$

  - Typically it ranges between -1 and 1.
  - The closer to 1 the better.

- We can calculate the Average Silhouette width for a cluster or a clustering solution

Internal Measures: *Connectivity*

- *Connectivity* captures the degree to which objects are connected within a cluster by keeping track of whether the neighboring objects are put into the same cluster

- For each object i
  - The closets k neighbors are found and ranked w.r.t. their similarity to i
  - Calculate $s_i = j_1 + j_2 + \ldots + j_k$, where j_r (r=1, 2, …, k) is zero if object i and the corresponding neighbor are in the same cluster and $1/r$ otherwise.
  - The connectivity score for the considered clustering solution of a set of n objects is then given by

    Connectivity $= s_1 + s_2 + \ldots + s_n$

  - Typically it ranges between 0 and infinity.
  - The closer to 0 the better.

Internal Measures: *Figure of Merit*

- Figure of merit (FOM) estimates the predictive power of clustering algorithms by assessing the quality of clustering results
- The predictive power is estimated by removing one feature from the data, clustering data based on the remaining features, and measuring the within-similarity of data values in the removed feature
- The FOM is the root mean square deviation in the left-out feature of the individual instances relative to their cluster mean
- This approach is analogous to *leave-one-out cross-validation*, which is a common predictive performance estimation method in supervised learning

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